

FINAL YEAR PROJECT PRESENTATION



Indian Career Stream Guidance & Sarkari Examination Gateway

PRESENTED BY

[Your Name / Team Members]

DEPARTMENT

Computer Science & Engineering / IT

GUIDED BY

[Project Supervisor / Guide Name]

TECHNOLOGY STACK

React 19 • Vite 8 • PHP 8.2 • MongoDB 8.0

Problem Statement & Background

The Aspirant's Challenge

- ✓ **Severe Stream Confusion:** Millions of students completing 10th, 12th, ITI, Polytechnic, Agriculture, Law, or Nursing do not know which government recruitments match their degree.
- ✓ **Application Fee Loss:** Misunderstanding age relaxation rules (OBC/SC/ST) and final-year appearing norms leads to rejected forms.
- ✓ **Neglected Non-Traditional Streams:** Vocational ITI trades, polytechnic diplomas, and agricultural graduates lack dedicated guidance.

Existing Portal Flaws (e.g. Sarkari Result)

- ✓ **Excessive Advertisement Clutter:** 10+ spam banners, popups, and clickbait download links per page.
- ✓ **Zero Academic Filtering:** Notifications are dumped in flat chronological lists without academic stream filtering.
- ✓ **No High-Yield Booklists:** Aspirants are left searching on YouTube/Telegram for syllabus weights and standard Indian books.

Comparison with Existing Systems

Evaluation Feature	Legacy Portals (Sarkari Result, FreeJobAlert)	Proposed System: KY EDU
User Experience	Cluttered, flashing banner ads, slow loading	100% Ad-Free Flipkart-style clean white UI
Stream Filtering	Unavailable (flat lists only)	12 Academic Streams (PCM, Agriculture, ITI, Law)
Board Categories	Generic (Jobs, Results)	10 Commission Boards (UPSC, SSC, RRB, AIIMS)
Eligibility Engine	Manual reading of 80-page notification PDFs	Interactive Multi-Factor Calculator
Preparation Guidance	None	Standard Indian Booklists & Weightage Breakdown
Administrative Security	Unauthenticated forms	Token & PIN Protected Admin Portal

Key Objectives of KY EDU

1. Stream Discovery

Connect candidates to verified exams based strictly on academic streams (10th, 12th PCM, Commerce, Agriculture, ITI, B.Tech, Law, Nursing).

2. Real-Time Search

Instant omnibar with live auto-suggestions and tokenized search matching titles, posts, qualifications, and pay scales without false negatives.

3. Eligibility Calculator

Auto-evaluate candidate qualification, age ceilings, reservation relaxations (OBC/SC/ST/PwD), and final-year appearing rules.

4. Standard Booklists

Equip aspirants with authoritative textbooks (M. Laxmikanth, RS Aggarwal, Spectrum) and high-yield scoring chapters.

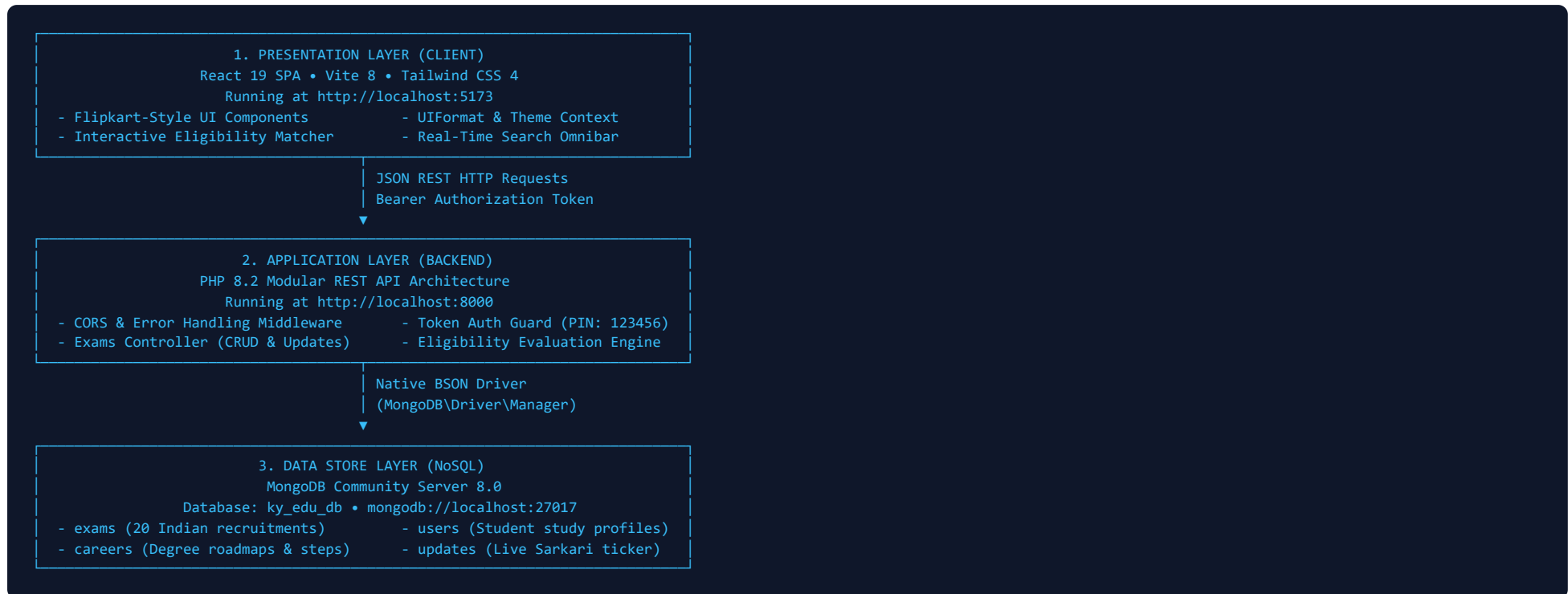
5. Protected Admin

Isolate exam updates and circular broadcasting behind multi-factor PIN (123456) and Bearer token authentication.

6. Multi-View Formats

Offer 4 view layouts: Cards Grid (Flipkart Style), Sarkari Table, Fast Scanner, and Stage Pipeline Timeline.

Three-Tier System Architecture



Hardware & Software Environment

Software Stack

- ✓ **Operating System:** Windows 10 / 11 / Linux
- ✓ **Frontend Framework:** React 19, Vite 8, Tailwind CSS, Lucide Icons
- ✓ **Backend Engine:** PHP 8.2 with native php_mongodb driver
- ✓ **Database:** MongoDB Community Server 8.0 (NoSQL)
- ✓ **Runtimes:** Node.js (v20+ LTS) & PHP Built-in Server
- ✓ **Development IDE:** Visual Studio Code

Hardware Environment

- ✓ **Processor:** Intel Core i3/i5 or AMD Ryzen 3/5
- ✓ **RAM:** 4 GB Minimum (8 GB Recommended)
- ✓ **Disk Storage:** 10 GB SSD / HDD available space
- ✓ **Display:** 1366×768 or 1920×1080 resolution
- ✓ **Network Ports:** Port 5173 (Frontend) & Port 8000 (Backend)

Student / Aspirant Panel Modules

Smart Omnibar

Live search with auto-suggestions dropdown showing exam title, category, and total posts in real-time as you type.

Dual Filter Chips

Flipkart Blue (#2874f0) for 10 Commission Boards and Vibrant Orange (#fb641b) for 12 Academic Streams.

Booklists & Syllabus

Detailed weightage percentages, high-yield chapters, and recommended textbooks by Indian authors.

Eligibility Matcher

14 recognized Indian qualifications, passing status, percentage, and category relaxation evaluation.

Career Roadmaps

Step-by-step career path roadmaps for ITI, Polytechnic, Agriculture, Law, and Civil Services.

Student Profile

Saves student standard and department for one-click personalized "Matched Sarkari Exams" filtering.

Protected Admin Portal Modules

1. Multi-Tier Authentication

- ✓ Quick 6-Digit PIN access (123456) for rapid access.
- ✓ SuperAdmin credentials (admin@kyedu.in).
- ✓ Automatic generation of Bearer tokens for all CRUD requests.
- ✓ Unauthorized mutating calls return 403 Forbidden.

2. Executive Analytics Dashboard

- ✓ Live counters for Active Exams, Pathways, and Live Circulars.
- ✓ Student department breakdown distribution chart.
- ✓ Recent activity audit feed for notices and exams.

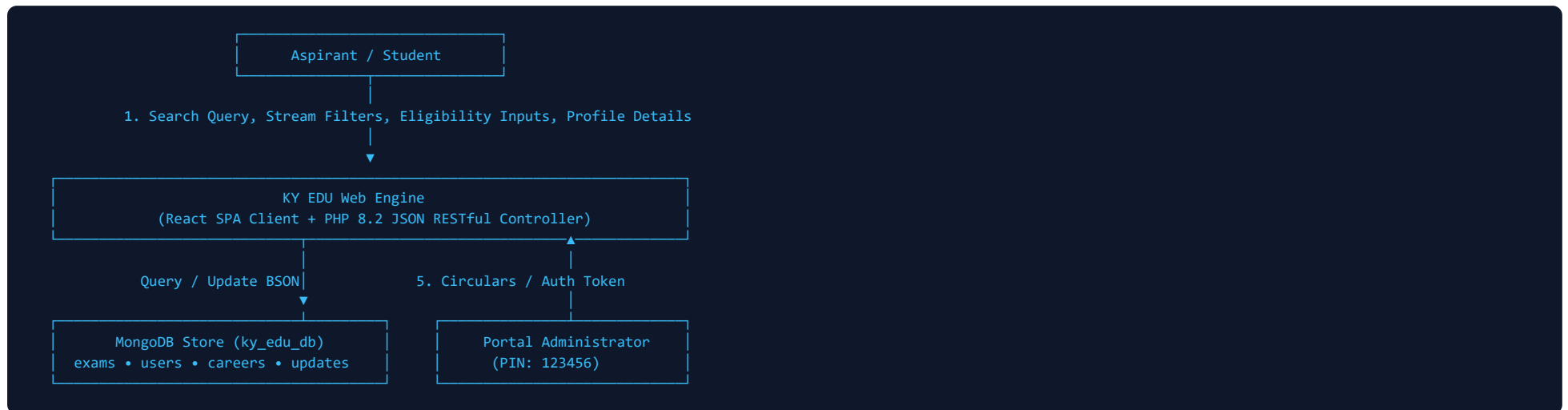
3. Full CRUD Management

- ✓ Create new exams with multi-stage exam tiers and salaries.
- ✓ Modify existing posts, syllabus topics, and eligibility.
- ✓ Delete outdated or expired notifications from MongoDB.

4. Live Notice Ticker Broadcaster

- ✓ Broadcast time-sensitive circulars (Admit Cards, Results).
- ✓ Directly updates the amber announcement ticker on the homepage.
- ✓ Export registered student data to CSV format.

Data Flow & Actor Interactions (DFD)



MongoDB Document Schema & Collections

📁 `exams` Collection (20 Documents)

`_id, title, shortName, conductingBody, category, stream, vacancies, posts[], eligibility{}, salary{}, stages[], subjectDetails[], timeline[]`

📁 `users` Collection (Student Profiles)

`_id, name, email, password, age, std (10th/12th/B.Tech), department, createdAt`

📁 `careers` Collection (Degree Roadmaps)

`_id, stream, title, overview, popularCourses[], topRoles[], averageStartingSalary, roadmapSteps[]`

📁 `updates` Collection (Live Circulars)

`_id, title, date, examShortName, badge, summary, link, createdAt`

20 Authentic Indian Recruitments Covered

Educational Stream	Examinations Populated in MongoDB
10th Pass / ITI Trades	SSC MTS (10th Pass) • RRC Group D (10th/ITI) • RRB ALP & ITI Tech
12th Pass (Any Stream / PCM)	SSC CHSL (10+2) • UPSC NDA (10+2 PCM) • State Police (12th/Grad)
Commerce & Banking / Finance	IBPS / SBI PO • SEBI Grade A (Finance/Law) • SSC CGL (Auditor)
Teaching & Education	CTET (Central Teacher Eligibility Test - B.Ed / D.El.Ed)
Computer Science & IT	ISRO Scientist / Engineer (CS/IT) • NIMCET (MCA Entrance)
Technical & Engineering	SSC Junior Engineer (JE) • RRB ALP & ITI Tech
Medical, Nursing & Pharmacy	AIIMS NORCET (Nursing Officer Level 7) • ESIC Pharmacist
Agriculture & Allied	IBPS SO AFO (Agricultural Field Officer)
Law & Judiciary	State Judicial Service (PCS-J) • SEBI Grade A (Law)

Testing & Verification Results

Test ID	Feature Tested	Test Input / Action	Observed Outcome	Status
TC-01	Public REST Read API	GET /api/exams.php	Returns 20 exams in JSON format with status 200	PASS
TC-02	Anti-Tampering Security	Unauthenticated POST /api/exams.php	Blocked with 403 Forbidden (authRequired: true)	PASS
TC-03	Admin Quick PIN	Submit PIN 123456	Admin session verified, Bearer token issued	PASS
TC-04	Smart Search Acronym	Query "nda" in Omnibar	Matches only UPSC NDA, eliminates false substrings	PASS
TC-05	Live Circular Ticker	Post new notice via Admin	Notice appears immediately on homepage banner	PASS
TC-06	Production Build Speed	npm run build via Vite 8	Compiles cleanly in 340ms with 0 errors	PASS

Key Advantages & Social Impact

1. 100% Ad-Free Learning Canvas

Eliminates the deceptive popups, malware redirects, and ad noise prevalent on legacy recruitment websites.

2. Empowering Non-Traditional Streams

First dedicated portal to prioritize ITI Craftsmen, Polytechnic Diploma holders, and B.Sc Agriculture graduates alongside Civil Services.

3. Financial Transparency

Displays clear 7th CPC Pay Matrices, starting in-hand salaries, and official government perks upfront.

4. Free Reference Booklists

Democratizes high-quality book recommendations (Laxmikanth, RS Aggarwal, Spectrum) for tier-2 and rural aspirants without coaching.

Future Scope & Enhancements

AI Mock Test Engine

Automatic generation of topic-wise practice quizzes based on high-yield syllabus percentages.

Mobile Application

Cross-platform React Native iOS and Android app with instant push notifications for deadlines.

Regional Languages

Full vernacular support in Hindi, Marathi, Gujarati, Bengali, and Tamil for rural aspirants.

WhatsApp Alert Bot

Direct circular notifications and admit card links sent automatically to candidate mobile numbers.

Commission Webhook Sync

Automated real-time scraping and synchronization with official UPSC, SSC, and NTA gazettes.

Aspirant Discussion Forum

Peer-to-peer Q&A community for strategy sharing and syllabus guidance.

Conclusion

Summary of Accomplishments

- ✓ Successfully architected and deployed **KY EDU**, a comprehensive Indian career and recruitment gateway.
- ✓ Integrated modern technologies: **React 19 SPA**, **Tailwind CSS**, **PHP 8.2 REST API**, and **MongoDB 8.0 NoSQL**.
- ✓ Solved major student pain points: ad clutter, lack of stream filtering, inaccurate eligibility, and missing book recommendations.
- ✓ Engineered an isolated, token-protected Admin Portal with 6-digit PIN login and real-time notice broadcasting.

Final Takeaway: KY EDU transforms how students discover government opportunities, empowering millions of Indian aspirants with transparent, verified, and personalized educational guidance.

PROJECT VIVA & EVALUATION

THANK YOU!

Questions & Discussion

PROJECT PORTAL

<http://localhost:5173>

ADMIN QUICK PIN

123456

BACKEND REST API

<http://localhost:8000>

DATABASE

MongoDB (`ky\_edu\_db`)